



Agent One — Technical Stack Documentation

Comprehensive Architecture, Framework, Model, and Infrastructure Reference

Version: 2.0.0 | Production Release

Live Deployment: <https://agent-one-rosy.vercel.app/>

Repository: <https://github.com/balrajpranay/agent-one>



Executive Summary

Agent One is an autonomous multimodal document intelligence workspace engineered to unify heterogeneous document ingestion, silent semantic domain routing, pixel-accurate spatial evidence grounding, and hybrid GraphRAG retrieval into a cohesive, high-performance platform.

This document serves as the **authoritative technical documentation** of all languages, frameworks, AI models, vector stores, graph engines, protocols, and infrastructure powering the system.



Architectural Topology

```
`` mermaid
graph TD
    subgraph ClientLayer ["🖥️ Frontend & Workspace HUD"]
        NextApp["Next.js 16 (App Router + RSC)"]
        ReactUI["React 19 + Tailwind CSS v4"]
        Viewer["Dual-Pane Spatial Bounding-Box Viewer"]
        GraphViz["Force-Directed SVG Graph Visualizer"]
        Theme["Dual-Theme Engine (Obsidian Dark / Clean White)"]
    end

    subgraph IngestionLayer ["📁 Universal Ingestion Engine"]
        PDF["pdf-parse (PDF AST & Coordinates)"]
        DOCX["mammoth (Word DOCX AST)"]
        MD["Markdown AST Parser"]
        Sheets["Tabular Parser (XLSX / CSV)"]
        OCR["Gemini 2.0 Multimodal OCR"]
    end
```

```
end

subgraph MCPLayer["🔒 Isolated Gemini MCP Server (mcp/gemini-server)"]
  Guard["Boundary Enforcement Tag Isolation"]
  ZodVal["Zod 4.6.5 Runtime Schema Validation"]
  MCPTools["6 Specialized MCP Analysis Tools"]
end

subgraph AIModelLayer["🧠 AI Reasoning & Foundation Models"]
  GeminiFlash["Google Gemini 2.0 Flash (Primary)"]
  GeminiPro["Google Gemini 2.0 Pro (Complex Reasoning)"]
  Classifier["10-Category Semantic Intent Router"]
  FallbackCascade["Multi-Model Auto-Failover Cascade"]
end

subgraph DataRetrievalLayer["📖 Dual-Retrieval & Knowledge Layer"]
  Neo4j["Neo4j Aura Cloud (GraphRAG + Leiden Clustering)"]
  Qdrant["Qdrant Cloud (384-dim Dense Vector Search)"]
  Supabase["Supabase (PostgreSQL + Cloud Storage + Auth)"]
  MemFallback["In-Memory Graph & Cosine Store (100% Offline Resilience)"]
end

subgraph DevOpsLayer["🚀 Infrastructure & CI/CD"]
  Vercel["Vercel Edge Global CDN"]
  Docker["Multi-Stage Production Dockerfile"]
  GitHub["GitHub Webhooks Automated CI/CD"]
end

ClientLayer --> IngestionLayer
IngestionLayer --> MCPLayer
MCPLayer --> AIModelLayer
AIModelLayer --> DataRetrievalLayer
DataRetrievalLayer --> ClientLayer
DevOpsLayer --> ClientLayer
```



Detailed Layer-by-Layer Breakdown

1. Frontend & Client HUD Architecture

The client tier is built on the bleeding edge of the React ecosystem, optimized for sub-100ms render speeds and instant visual feedback.

Component	Technology	Version	Architectural Role
Framework	Next.js (App Router)	16.3.1	Turbopack engine, React Server Components (RSC), optimized Edge routes, and streaming responses.
UI Library	React	19.2.8	Concurrent rendering, reactive UI state machines, and synchronized dual-pane inspection.
Static Typing	TypeScript	^5.0	Strict end-to-end type safety, generic schema enforcement, and zero any tolerance.
Styling & Design Tokens	Tailwind CSS	v4.0	Custom Obsidian (#090a0f) and Electric Emerald (#00FF85) design system.
Vector Iconography	Lucide React	1.31.0	Tree-shaken SVG vector HUD icons across all workspace views.
Data Visualization	Recharts	3.10.1	Interactive financial bar charts, liability exposure breakdown, and risk area charts.
Canvas Particle Engine	HTML5 Canvas	Native	High-performance interactive background dot-matrix visualization and coordinate crosshair overlays.
Theme System	Custom Theme Context	Custom	Instant zero-flicker toggle between Obsidian Dark and Clean White themes (themeContext.tsx).
Micro-Interactions	Canvas Confetti	1.9.4	Milestone and successful audit completion celebrations.

2. AI Reasoning, Vision & Foundation Models

Agent One utilizes Google's latest multimodal models with autonomous prompt engineering and fallback cascades.

Component	Model / Engine	Implementation	Purpose & Highlights
Primary Multimodal LLM	Google Gemini 2.0 Flash	@google/generative-ai 0.24.1	Ultra-fast token generation, cross-modal document comprehension, and low-latency structured extraction.
Complex Reasoning LLM	Google Gemini 2.0 Pro	Dynamic Auto-Switch	Deep cross-clause synthesis, multi-hop legal risk discovery, and dense tabular reasoning.
Multimodal Vision OCR	Gemini Vision Engine	Native Base64 / Stream	High-accuracy text and table OCR on scanned receipts, physical contracts, and diagrams.
Spatial Grounding	Normalized Bounding Boxes	[x, y, w, h] (0.0 to 1.0)	Maps extracted facts to exact visual coordinates on the original page, powering interactive crosshairs.
10-Domain Intent Classifier	Hybrid Rule + LLM Engine	src/lib/docClassifier.ts	Silently identifies 10 categories (Legal, Finance, Business, Insurance, Academic, etc.) without prompt bloat.
Model Fallback Cascade	Resilient Multi-Model Failover	src/lib/geminiClient.ts	Auto-failover sequence (gemini-2.0-flash → gemini-flash-latest → Pro) with 8s per-attempt timeout.

3. Knowledge Graph & GraphRAG Retrieval

To eliminate blind LLM hallucinations, Agent One constructs and queries a live knowledge graph.

Component	Technology	Version	Purpose & Implementation
Graph Database	Neo4j Aura Cloud	neo4j-driver 6.2.0	Fully managed cloud graph database for cross-entity relationship modeling.
Query Engine	Cypher Query Language	Native Cypher	Idempotent MERGE transactions modeling relationships: <code>DEFINED_IN</code> , <code>CARRIES_RISK</code> , <code>DUE_ON</code> , <code>OBLIGATED_TO</code> .
Graph Transformation	Graph Transformer	src/lib/graphTransformer.ts	Ingests raw document analysis and converts entities/metrics into a typed node-edge graph.
Community Detection	Leiden Algorithm	Graph Clustering	Detects thematic document communities to expose hidden organizational risks and dependency cycles.
Offline Graph Fallback	In-Memory Graph Engine	src/lib/graphRag.ts	High-speed JavaScript Map/Set graph store providing 100% offline functionality if Neo4j is offline.

4. Vector Search & Semantic Chunk Retrieval

Component	Technology	Specification	Implementation Detail
Vector Database	Qdrant Cloud	REST Client API	High-scale vector similarity engine with cosine metric indexing in collection docfin_documents .
Dense Embeddings	Sentence-Transformers	all-MiniLM-L6-v2	384-dimensional dense semantic embeddings generated via HuggingFace Inference API.
Local Vector Fallback	Deterministic Cosine Generator	384-dim Float Vectors	Normalized local vector generation guaranteeing vector retrieval continuity during API outages.

5. Model Context Protocol (MCP) Server Architecture

Agent One isolates LLM operations behind a dedicated, hardened MCP server located in mcp/gemini-server .

`text`

Boundary Guard Isolation Format:

- <<>> - System instructions and security policy
- <<>> - Activated domain-specific intelligence rules (e.g. legal.md, finance.md)
- <<>> - End-user query and conversational history
- <<>> - Grounded document text, tables, and bounding boxes
- <<>> - Schema-validated MCP tool invocation payloads

MCP Tool Name	Schema Validation	Purpose
<code>gemini.analyzeMultimodal</code>	Zod 4.6.5	Ingests document buffers/images and produces full structural analysis.
<code>gemini.extractStructured</code>	Zod 4.6.5	Extracts typed entities, dates, liability clauses, and numerical metrics.
<code>gemini.reason</code>	Zod 4.6.5	Performs multi-hop logical deductions across document sections.
<code>gemini.answerWithEvidence</code>	Zod 4.6.5	Answers queries with verifiable citations and spatial coordinate bounds.
<code>gemini.compareDocuments</code>	Zod 4.6.5	Computes semantic diff between revisions (added, removed, shifted figures).
<code>gemini.embedText</code>	Zod 4.6.5	Generates normalized dense embeddings for vector indexing.

6. Universal Document Ingestion & Parsers

The ingestion subsystem automatically identifies MIME types and parses diverse business file formats:

Format / Extension	Parsing Engine	Package	Key Output Deliverables
PDF (.pdf)	pdf-parse	2.4.5	Layout preservation, page-by-page text streaming, coordinate anchor mapping.
Word (.docx, .doc)	mammoth	1.13.0	Hierarchical heading extraction, table cell preservation, bullet points, and raw AST.
Markdown / Text (.md, .txt)	Native AST	Native	Code block identification, markdown table parsing, nested section hierarchy.
Spreadsheets (.xlsx, .csv)	Tabular Ingestor	Native	Columnar data normalization, transaction log structuring, Recharts data binding.
Scanned Documents (.png, .jpg, .jpeg)	Gemini Vision OCR	Multimodal Flash	Pixel-level OCR, document orientation correction, spatial bounding box detection.

7. Authentication, Multi-Tenancy & Persistence

Service / Tool	Library	Purpose
Supabase Cloud	@supabase/supabase-js 2.109.0	Managed PostgreSQL database for audit trails, document catalogues, and storage buckets.
Google OAuth 2.0	@react-oauth/google 0.13.5	Real one-click enterprise authentication with Google Client ID & Secret verification.
Token Verification	jwt-decode 4.0.0	Client-side decoding and session expiration handling.
Multi-Tenant Isolation	Custom Middleware	Strict data boundaries ensuring users never access cross-tenant documents (10/10 automated tests pass).

8. DevOps, CI/CD & Cloud Infrastructure

Infrastructure Layer	Platform / Tool	Configuration Details
Production Hosting	Vercel Edge Global CDN	Serverless Next.js deployment at https://agent-one-rosy.vercel.app/ .
Automated CI/CD	GitHub Webhook Integration	Zero-downtime automated deployment triggered instantly on every git push origin main .
Containerization	Docker	Multi-stage production Dockerfile with lightweight Alpine Node.js runtime and docker-compose.yml .
Package Management	npm	Deterministic dependency tree locked via package-lock.json .
Testing Harness	Custom Test Suite	7-stage automated verification (scripts/test-runner.mjs) covering types, lint, pipeline, isolation, and builds.

Environment Variables Reference Matrix

All 13 environment variables configured for production on Vercel:

Variable Name	Environment	Sensitivity	Fallback Strategy	Purpose
GEMINI_API_KEY	Production / Local	Secret	Required	Google Gemini 2.0 Flash/Pro inference, OCR, and reasoning.
GRAPH_RAG_ENABLED	Production / Local	Config	Defaults to true	Enables Neo4j knowledge graph integration.
NEO4J_URI	Production / Local	Config	In-Memory Graph	Bolt/Neo4j+s endpoint URL for Neo4j Aura cloud instance.
NEO4J_USERNAME	Production / Local	Config	In-Memory Graph	Database user authentication credential.
NEO4J_PASSWORD	Production / Local	Secret	In-Memory Graph	Secure password for Neo4j database authentication.
QDRANT_URL	Production / Local	Config	In-Memory Vectors	REST endpoint for Qdrant Cloud cluster.
QDRANT_API_KEY	Production / Local	Secret	In-Memory Vectors	API key for authenticated Qdrant operations.
NEXT_PUBLIC_SUPABASE_URL	Production / Local	Public Config	In-Memory Store	Supabase project REST and API gateway URL.
NEXT_PUBLIC_SUPABASE_ANON_KEY	Production / Local	Public Key	In-Memory Store	Supabase public client anonymous authentication key.
SUPABASE_SERVICE_ROLE_KEY	Production / Local	Secret	In-Memory Store	Privileged Supabase backend administrative access key.
NEXT_PUBLIC_SUPABASE_BUCKET_NAME	Production / Local	Public Config	Defaults documents	Storage bucket name for uploaded raw files.
NEXT_PUBLIC_GOOGLE_CLIENT_ID	Production / Local	Public Config	Mock Auth	Google OAuth 2.0 Web Client identifier.
GOOGLE_CLIENT_SECRET	Production / Local	Secret	Mock Auth	Google OAuth 2.0 backend server secret.



Verification & Audit Commands

To verify the entire tech stack locally:



```
` bash
```



1. Full 7-stage automated production verification

```
npm test
```

2. TypeScript compilation type-check

```
npx tsc --noEmit
```

3. Universal ingestion pipeline test

```
npx tsx scripts/test-universal-pipeline.mjs
```

4. Multi-tenant user data isolation test (10/10 passing)

```
node scripts/test-user-isolation.mjs
```

5. Production Next.js Turbopack build

```
npm run build
```



```
``
```

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